MILITARY SPECIFICATION

TABLE, SCRIBING, TRACING, DRAFTING:

FLIP TOP

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a combination flip-top scribing, tracing, and drafting table for use in a mobile van application.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-D-635	- Drawing Board and Drawing Board Trestles.
QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Packaging, Methods of.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The table shall be as shown in figures 1 and 2, and as specified herein. The table shall consist of a frame, drawing board-light box, top and drawer module. The drawer module shall swing clear of the pivoting work surfaces. The drawing board-light box top shall be a pivoting, flip-flop action type for ready conversion and security in either a work or mobile transport mode.

3.2 First article (first-produced table). The contractor shall furnish one table for examination and testing within the time frame specified (see 6.2), to prove that his production methods and choice of design detail will produce tables that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Frame. The frame for the table shall be constructed of square steel tubing conforming to ASTM A512, A513 or A519 with a square cross-sectional stock not less than 2 inches. Joints shall be welded or double bolted to assure stability and ready assembly. Overall dimensions shall be not more than 47 inches wide, approximately 34 inches deep, and approximately 42 inches high. Each leg shall have a locking level device independent of tie-downs required for mobile application.

3.5 Drawing board-light box top. The top shall be a combination drafting board-light box design with a back-to-back construction, pivot-point and trunion bracket on each side for a flip-flop pivoting action, and adjustable locking stops at fixed positions of a elevation to the horizontal. The elevations shall be approximately 5 and 10 degrees. The front edge of both the drawing board and light box shall be provided with drawing guards. The left and right edges shall have a metal strip to accommodate the use of a T-square.

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3.5.1 Drawing board. The drawing board shall have a one-sided working surface and shall be approximately 42 inches wide, approximately 31 inches deep, and approximately 1 inch thick. The material and construction of the board shall be in accordance with GG-D-635.

3.5.2 Light box. The light box shall have a working surface not less than 30 inches long and 30 inches wide. The box shall be deep enough to house four 30-inch fluorescent fixtures, ballasts and lamps. Supports, lamps, fixtures, and reflecting surfaces shall be arranged for diffusion of uniform stable lighting. The glass plate shall be a safety glass, secured to the light box for flip-top action, removeable for maintenance and cleaning, and mounted to prevent breakage in both the work and the transport mode, which is with the glass in the bottom positions.

3.5.2.1 Light diffusion. Light diffusion shall be accomplished by any of the following methods:

(a) Surface of glass fine-grained.
(b) Opal glass.
(c) Opaline plastic sheet.

3.6 Drawer module. The table shall be fitted with a two drawer module that will swing forward out of the path of the flip-top action of the table top, around the right front leg of the frame. The module shall be positioned by a spring-loaded clasp remotely released by a lever. The release lever shall be located on the upper right portion of the drawer module. The clasp action shall securely fasten and position the module to the side frame in a 3-point control with guide plates or rests. The drawer module shall be approximately 12 inches wide by approximately 26 inches deep by approximately 18 inches high and shall be an accessory to, but not a part of the table frame. The module shall also be lockable with a single key.

3.7 Tray. When specified (see 6.2), the tray shall be approximately 12 inches long by 7 inches wide. The height shall be a diagonal approximately 3 inches to 1-1/2 inch. The method of temporarily securing the tray to the drawer module shall be by magnets and clips (see figure 2).

3.8 Electrical system. The electrical system shall operate from 120-volt, 60 Hertz (Hz), single phase current.

3.8.1 Power cord, switches, and receptacles. Two electrical receptacles, a fuse and an on-off switch shall be mounted vertically on the top part of the leg. The power input cord shall be approximately nine (9) feet long.

3.9 Identification marking. Each table shall be identified in accordance with MIL-STD-130.

3.10 Treatment and painting. The table shall be cleaned, treated with a primer and painted in accordance with accepted industry practice. Unless otherwise specified (see 6.2), the color shall be light green.

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3.11 Workmanship. All parts, components, and assemblies of the table including castings, forgings, molded parts, stampings, bearings, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Nonfunctional sharp edges, projecting points and welding burrs shall be avoided. The glass working surface shall not be cracked, chipped, or broken. Drawers shall not bind and shall fit the drawer opening. Glue joints shall not have any looseness of wood pieces, wide gaps in seams, lack of rigidity or presence of warp. The paints shall not chip, streak, or flake.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsbile for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced table (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced table inspection.

4.3.1 Examination. The first produced table shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced table shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. The table shall be examined as specified herein for the following major defects:

101. Material not as specified.
102. Frame not as specified.
103. Drawing board-light box top not as specified.
104. Drawer module not as specified.
105. Tray not as specified.
106. Electrical system not as specified.
107. Identification not as specified.
108. Treating and painting not as specified.
109. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Pivoting (flip flop) conversion. With the table assembled and the drawing board-light box in the drawing board mode, test the working surface for squared edges, stability and elevation positions. Unlatch and swing the drawer module forward around the right forward leg. Release the elevation fixture and rotate the drawing board-light box on its pivots and trunions to the light box mode. The conversion shall be made without binding and with clearance of both frame and drawer module. After the drawer module has been swung into working position, it shall be tested for stability in its mounts and spring latch by pushing and pulling on the module. Evidence that the working surface edges are not square, or evidence of binding during pivoting, rotating and elevating, as evidence of instability or lack of clearance shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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110. Materials or containers not as specified. Each incorrect material or container shall constitute one defect.
111. Unpainted exterior metal surfaces requiring a contact preservative not coated as specified.
112. Lamps not secured as specified for level A.
113. Openings into electrical components not sealed as specified for level A.
114. Electrical cord not secured as specified for level A.
115. Table top and drawer section not secured as specified for level A.
116. Glass working surface not cushioned as specified for level A.
117. Table not covered with a shroud as specified for level A.
118. Blocking, bracing and anchoring not as specified for level A or level B.
119. Strapping not as specified for level A or level B.
120. Marking illegible, incorrect, incomplete or missing.

5. PACKAGING

5.1 Preproduction pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The preproduction pack may be prepared utilizing either the first-produced table or a production table. When the first-produced table is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the first-produced and production tables.

5.2 Preservation. Preservation shall be level A, or Commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unpainted metal surfaces. Unpainted exterior metal surfaces of the table requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with MIL-P-116.

5.2.1.2 Electrical components. The fluorescent lamps shall be secured in their sockets with tape conforming to PPP-T-60, type IV. Openings into electrical components such as switches and outlets shall be sealed with tape specified herein. The cord shall be coiled and the coil secured with not less than three evenly spaced ties of cotton tape or twine.

5.2.1.3 Table top and drawer section. The table top and drawer section shall be secured. The glass working surface shall be covered with minimum 1-inch-thick cushioning material conforming to PPP-C-843, type II, class A. The cushioning material shall be secured in place with tape conforming to PPP-T-60, type IV.

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5.2.1.4 Complete table. The complete table, with components preserved as specified herein, shall be covered with a shroud fabricated from barrier material conforming to PPP-B-1055, class C-1 or C-2. The shroud shall be secured with tape conforming to PPP-T-60, type IV.

5.2.2 Commercial. Each table shall be preserved in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be level A, level B, or Commercial, as specified (see 6.2).

5.3.1 Level A. Each complete table, preserved as specified in 5.2, shall be packed in a close-fitting box conforming to PPP-B-621, class 2, or PPP-B-601, overseas type. Blocking, bracing and anchoring of the table within the box shall be in accordance with MIL-STD-1186. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.3.2 Level B. Packing of tables shall be as specified in 5.3.1 for level A, except that boxes shall be domestic class or type and strapping shall be finish A.

5.3.3 Commercial. Tables, preserved as specified in 5.2, shall be packed in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The table is used in the Cartographic Image Base Products, and Military Information Subsystems of the Mobile Topographic Support System (TSS).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the first-produced table (see 3.2).
(c) When a tray is required (see 3.7).
(d) Color required if other than as specified (see 3.10).
(e) When a preproduction pack is not required (see 5.1).
(f) Time frame required for submission of the preproduction pack, when applicable (see 5.1).
(g) Degree of preservation and degree of packing required (see 5.2 and 5.3).
(h) When finish A strapping is required (see 5.3.1).

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6.3 First-produced table. Any change or deviation of production tables or packaging packs from the approved first-produced table or packaging pack will be subject to the approval of the contracting officer. Approval of the first-produced table or packaging pack will not relieve the contractor of his obligation to furnish tables or packaging packs conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

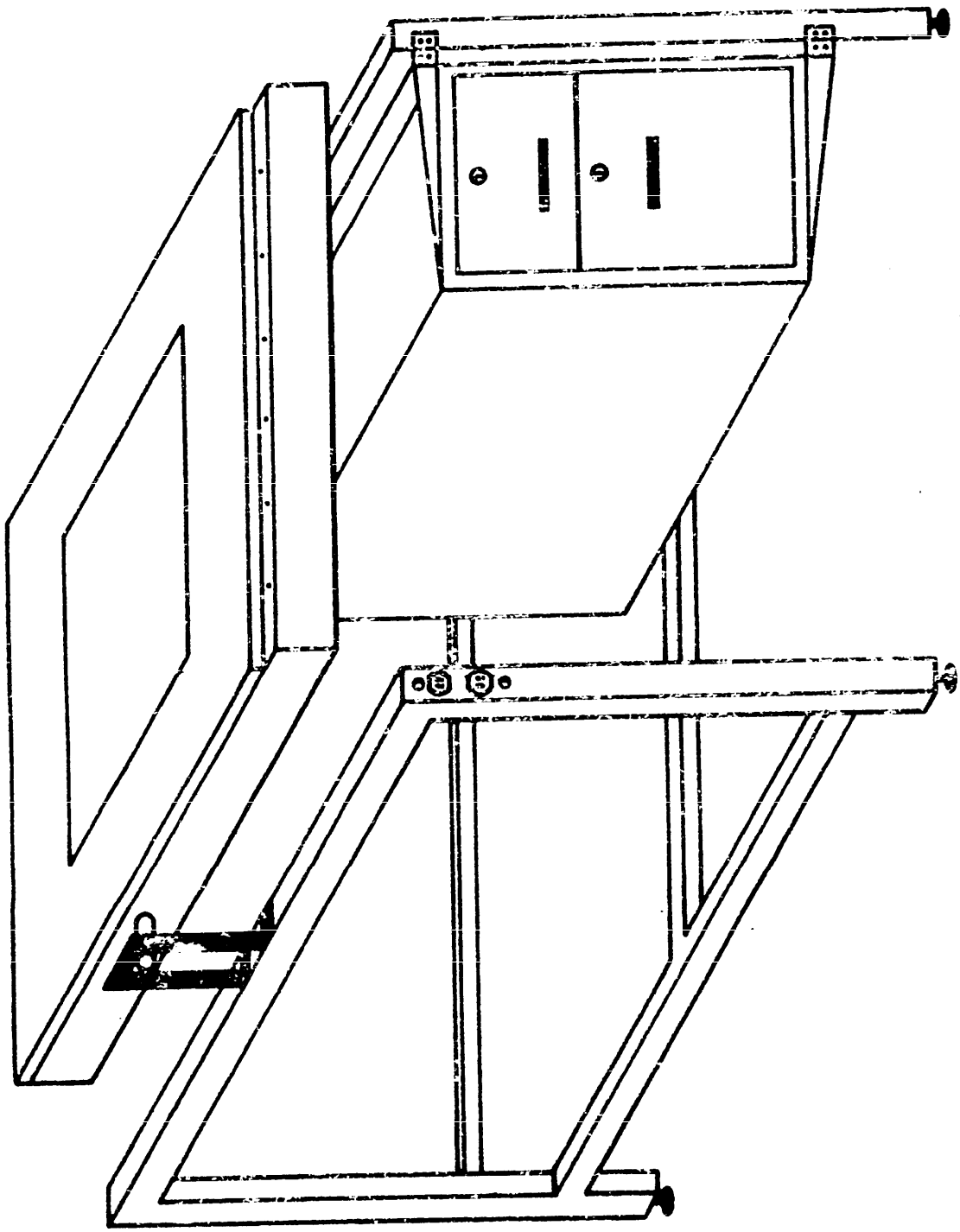
6.5 Information figures. Figure 1 and 2 shows a type of table and tray which has been found accpetable; however, the figures are included for illustration only and are not intended to preclude the furnishing of another table and tray which conforms to this specification.

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FIGURE 1. Table, scribing, tracing, drafting.

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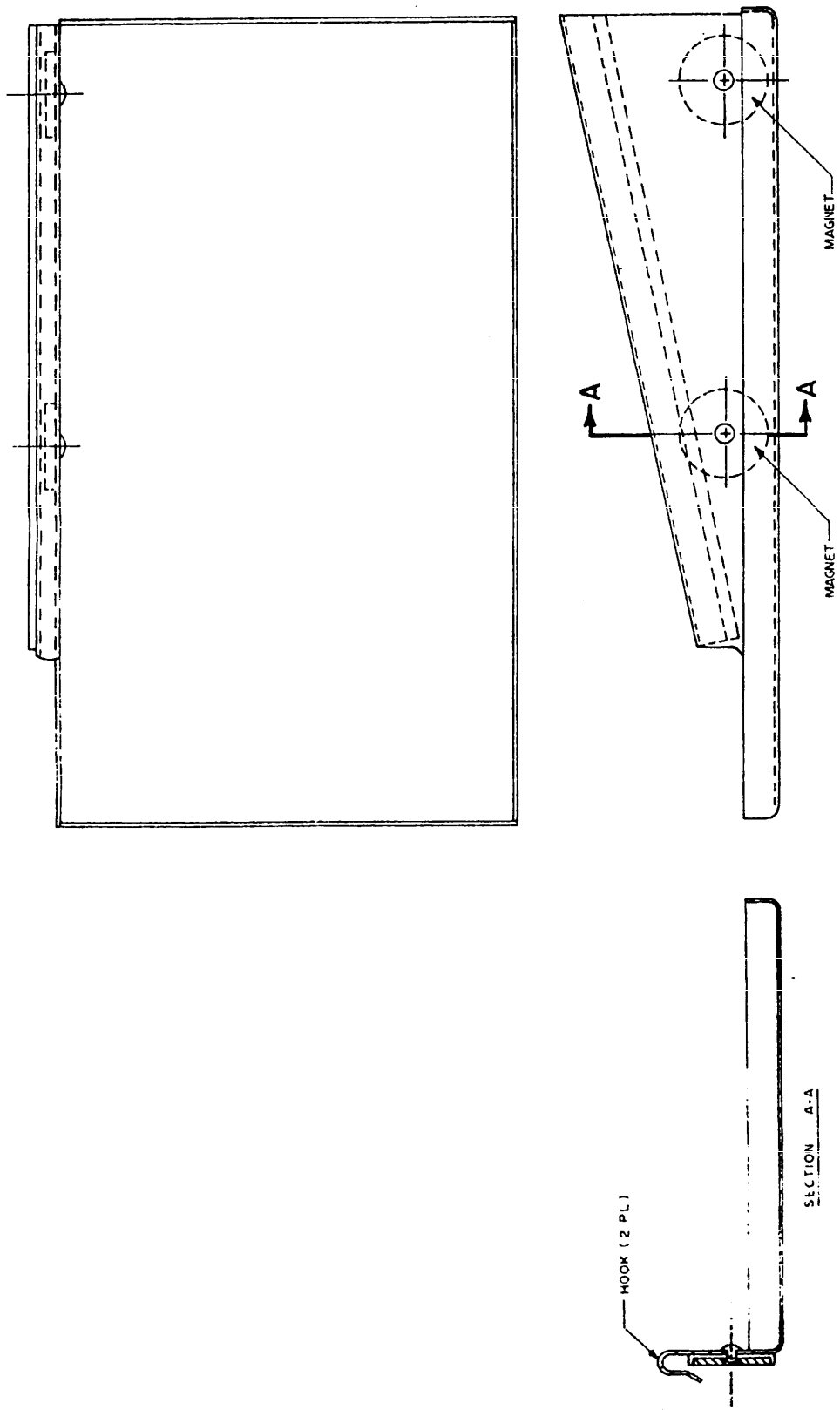


FIGURE 2. Tray.

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